RD900 locus structure project description

Background

We have previously published a paper detailing variation across the Mycobacterium tuberculosis Complex (MTBC) in the RD900 region. This region was first described as a lineage specific locus in M. africanum GM041182 strain (Bentley et al. 2012) and was thought to be deleted in M. bovis and "modern" M. tuberculosis lineages. RD900 region was not present in the original M. bovis AF2122/97 reference genome annotation, but was found to be actually present upon re-sequencing. These species show more than 99% genetic identity but exhibit distinct host preference and virulence. Differences in key locii are likely to be important in determining this.

The RD900 region comprises two homologous genes, pknH1 and pknH2, encoding a serine/threonine protein kinase PknH flanking the tbd2 gene. Our analysis revealed that RD900 has been independently lost in different MTBC lineages and different strains, resulting in the generation of a single pknH gene. Importantly, all the analysed M. bovis and M. caprae strains carry a conserved deletion within a proline rich-region of pknH, independent of the presence or absence of RD900. We hypothesized that deletion of pknH proline rich-region in M. bovis may affect PknH function, having a potential role in its virulence and evolutionary adaptation.

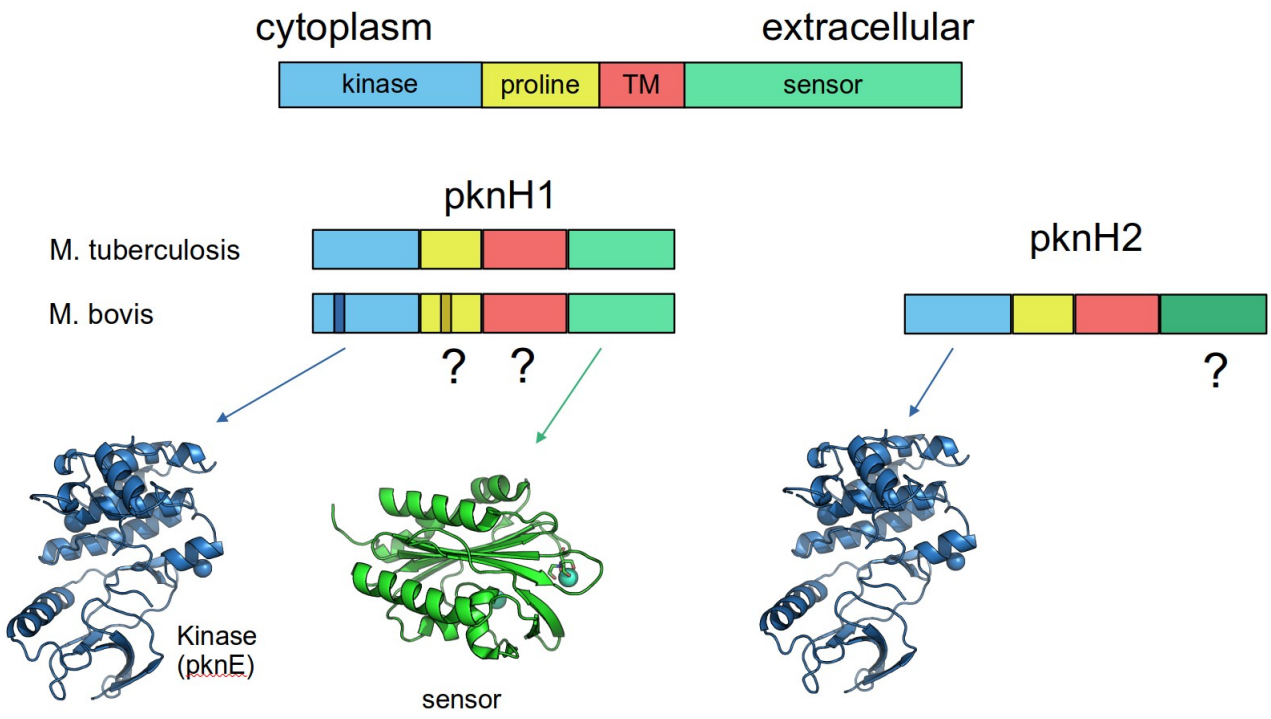


Figure 1: Structure of the pknH1 and pknH2 genes.

In this study we examined over 60 different MTBC strains representing lineages across the MTBC to see variations in the region. To accurately detect locus structure we used the raw read data instead of

relying on assembled genomes which may have errors due to homology in the kinase domains of the pknH proteins, if both are present. Raw fastq files for all strains were downloaded and aligned to the most complete version of the RD900 locus encoded in the Mycobacterium africanum (MAF) genome. This allowed missing features to be seen as gaps, as shown below.

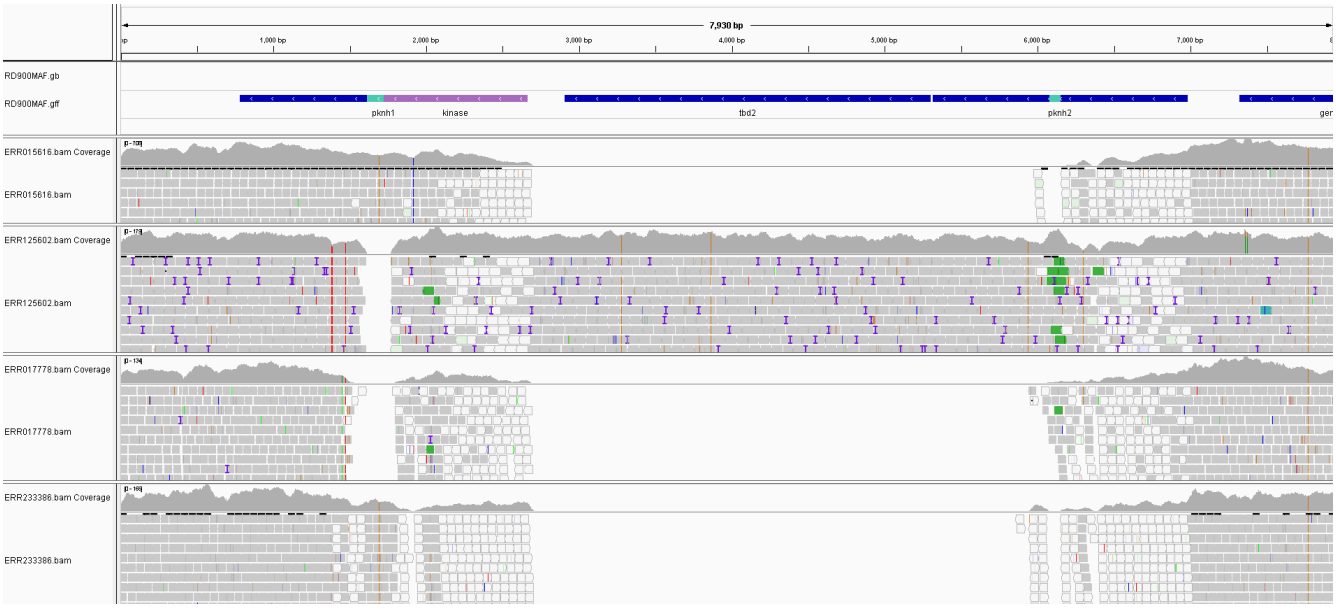


Figure 2: Alignments of some samples to the RD900 MAF region show the gaps where features are missing. The white reads are those multi-mapping to the homologous kinase domains of pknH1. This is what causes issues in assembly at this locus.

Objectives

Since this project was done in 2019, new data has been produced for all of these mycobacteria genomes including the animal adapted species we are most interested in. We will use this new data to provide a more complete analysis. As well as using raw read data as before we will use complete assemblies as a complementary approach. Complete assemblies should cover the whole locus and give a reliable picture of it's structure.

We will:

- Retrieve new raw read post 2019 data to apply the same approach as before
- We will find all complete assemblies for the same set of species. In particular this may help find a more complete locus for Africanum. which we can align to.
- In both cases we need to reduce redundancy drastically to get a manageable dataset. I will handle this part.
- It might be that we can perform both methods in parallel and then compare results but one will feed into the other.

Methods

1. The assembly approach

There are >600 compete MTBC assemblies and >120 incomplete long read assemblies available. These have already been downloaded. We actually do not need to reduce these now, as we can just extract the locus for all of them first and then reduce redundancy at the locus level.

Extracting the RD900 locus

We chose to adopt an anchor based extraction + alignment method. We find the conserved anchor areas around the locus first, then **BLAST the two flanking anchor genes separately** (not the internal content), then take whatever lies between their hit coordinates in each assembly as "the locus," regardless of what's in there. This sidesteps the paralog-confusion problem and should correctly extract the interval *whatever its content*: intact, deleted, partial, or bearing a large insertion, with no assumption about what should be inside.

Comments/caveats on this approach in general:

- Anchor choice matters more than anything else. They need to be genuinely unique and conserved across *all* species in - so it's worth explicitly checking copy number of your chosen anchor genes across the assemblies.
- Assembly fragmentation breaks it silently. But we are using complete genomes so that should be ok, but do not assume 100% extraction across all assemblies.

Annotate-cluster-reduce method for grouping locus sequences

Since we want to see all possible diversity in the locus across species but avoid duplication, we want to do the following:

1. annotate the protein structure for all sequences using our known proteins
2. cluster/group by structure
3. mash reduce within each group and get a representative sequence

We will also be interested in whether the same species is in >1 structure groups (should be the case for *M.bovis*).

On threshold: 0.001 is a good starting point for collapsing near-duplicate locus sequences. But use different thresholds to see the results. If we use very small thresholds we will get too many groups.

Find lineages

This will be done with a tool called TB-profiler. I will do this part and add it to the table.

Compare the RD900 sequences

We can do a multiple sequence alignment of the locus for all samples. Using mafft is fastest.

We can also build a phylogeny with *fasttree* to get an idea of how the locus sequences line up as compared to their species relationship.

To compare protein structure of the locus groups we can do it visually with Clinker or Python (see below).

Analysis steps

Note: when using Python scripts you can always run python <scriptname.py> -h to get help.

Here are some of the commands for the assembly analysis. Your filenames may differ in some cases.
Extract the up and downstream anchors:
Here is the link to the genome viewer for the reference Africanum genome where you can extract the flanking sequences from the region.
https://www.ncbi.nlm.nih.gov/nuccore/FR878060.1?report=graph&from=1411192&to=1416081

The script locus_extraction.py will then extract the locus sequences. Note that you can run the extraction on the complete and incomplete genomes separately and then combine the resulting fasta together (using cat). Or put all genomes in one folder if you prefer.

```
python locus_extraction.py --up upstream_anchor.fa --down downstream_anchor.fa --assemblies mtb_genomes
```

The script annotate.py will identify all the proteins in each locus and create a gbk file for each one in a folder.

```
python annotate.py -p RD900MAF.faa -o annot RD900_extracted.fa
```

To cluster the locus sequences after annotation we use this script:

```
python locus_cluster.py --input RD900_extracted.fa --gbk annot --threshold 0.0001
```

This will produce an output folder structure_results where there is also a structure summary table. This will be useful for curating the sequences we want to select ourselves if needed. It also writes out the representative gbk files to a folder called annot_repr and the sequences to a fasta file inside structure_results.

We can use clinker to view the output

```
clinker annot/*.gbk -o clinker_alignment.html -p clinker_plot.html -i 0.75
```

There is also a custom Python script to plot the locus comparison that will be more easy to customise later. This will plot whatever gbk files you give it. So you can only give it the representative ones from the structure results table for example.

```
python locus_plot.py --input RD900_extracted.fa --gbk annot --table asm_complete_mtbc.csv
```

Sequence alignment

Align the representative sequences:

```
mafft --auto --thread 4 RD900_extracted.fa > RD900_extracted_repr.fa
```

Make a phylogeny from the alignment:

```
fasttree -nt -gtr RD900_extracted_aligned.fa > RD900_locus_tree.nwk
```

2. Protein based analysis

This approach allows us to compare the protein sequences for specific proteins where they are present, especially pknH1 and 2.
We can extract the amino acid sequences and align them to see where the most variation is and where there are deletions/insertions. The amino acid sequences are already in the genbank annotations we made. Use mafft or similar to align them.
We may also predict the structure of these proteins and see if there is any predictable effects on structure.

Analysis steps

Write a Python script to extract all the sequences for a specific protein. Try pknH1 first but it should work for any of them.
Reduce the sequences as before with mash.
Align the proteins sequences and plot them.

3. Raw reads alignment approach

This is the method we used before. It avoids the issues of erroneous assembly which may exist even in the complete ones. Especially in regions like RD900. Also there is a larger raw read dataset (as not all raw WGS data is assembled and uploaded and most are never made into complete genomes). It remains to be seen how we can use this data if we have results from the assemblies (which we did not have before).

Datasets

On the shared drive there are now several sets of data:

mtbc_assemblies.zip: this zip file contains all ~680 complete assemblies.
mtbc_assemblies_lr.zip: this zip file contains incomplete long read assemblies.
asm_all_mtbc.csv: the list of details for all the assemblies. Includes the country. This will be useful later for identification.

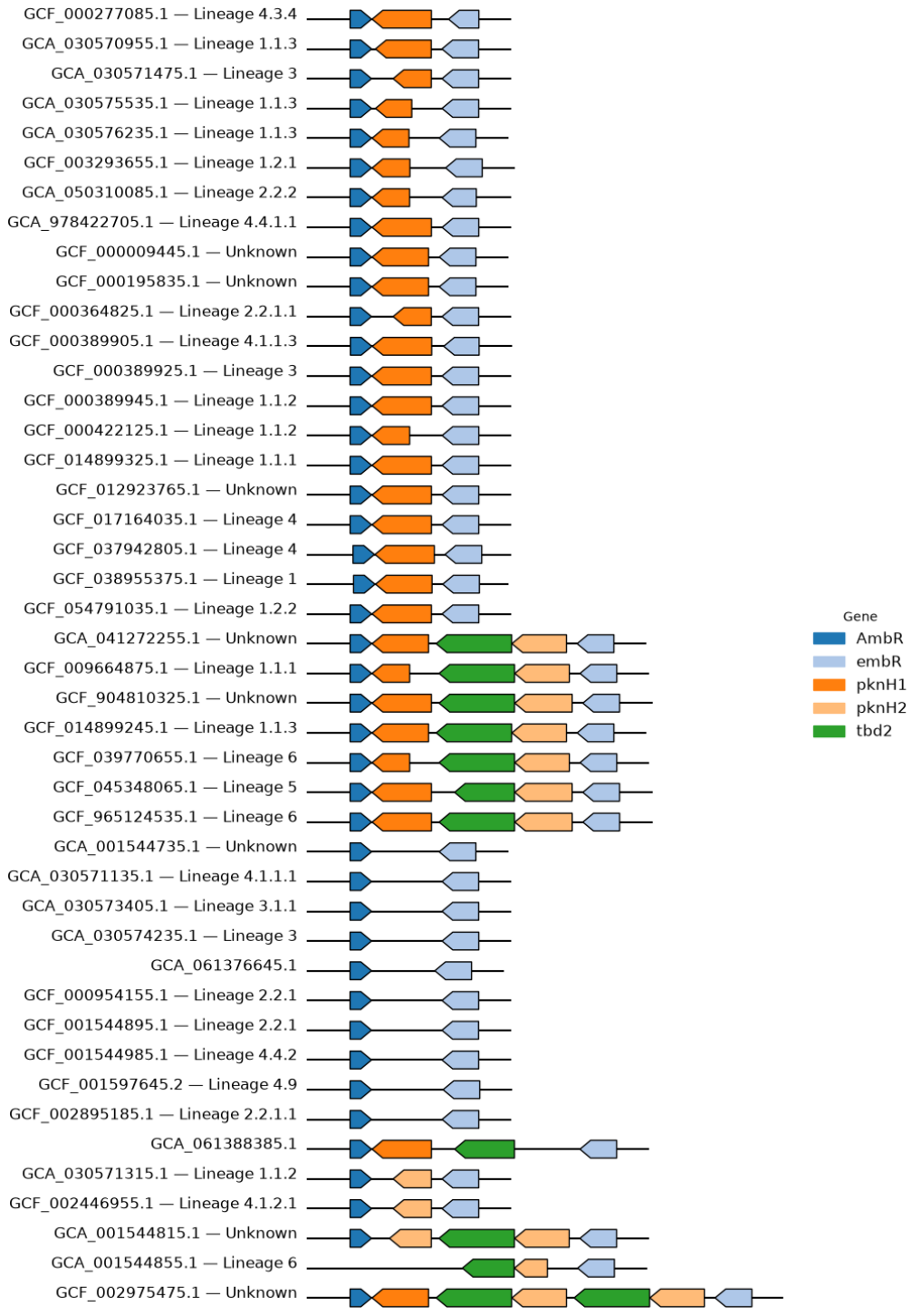
Software installation

On Ubuntu for windows (WSL) you can install most software with apt. e.g. sudo apt install bwa. Unfortunately a small number of bioinformatics tools, even those written in Python, are much easier to install with conda. Conda is a last resort if we can't install another way.
On macOS it may be installable with brew, if not conda may be needed.

Results

This section is for preliminary results and suggestions for further analysis.

Ultimately we will want one or two examples only of the locus structure for each species to see the configurations. You can use the structure_summary file to curate the representative gbk files into a separate folder for plotting.
Here is an example result. Note that this won't just spit out the right answer. It's meant to help us see the within and between lineage diversity. So you will have to play around with it.



Things to look for:

- There may be unusual outliers and they may be incorrect. Caused by wrong extraction? Are they real?
- Are the results we see consistent with the previous study?
- Does the same species appear in multiple structure groups?
- Should we use lower thresholds to see more subtle changes in the protein sequences like insertions.
- Possibly we could update the plotting code to show small changes or use sequence alignment?
- Is there a larger version of the M. Africanum locus than the one that we are currently using as a reference?

Next steps:

- How can we add the read data from the previous study, is it needed?